

Web Scraper

For doing well at any kind of real world goals, you need information and data. (Unless you can hear the voice of God maybe.. But even that's information from a source.) Some sources are better than others (*ahem*) but no good source will be complete by itself. Also, the way websites work, aggregating information from multiple pages cannot be done at near light speed by a human. (Unlike the cup stacking world championships.) It can be done by a machine though, machines you likely already possess. All that's remaining is to transcribe your wishes into the computer's command. And that's where python comes in. (Technically you could scrape in any language that can communicate over http but for the purposes of this FastTrack, no other language exists.) With python, just like a browser, you will request and retrieve the page. Unlike a browser, and depending on the module you use, the page will not be displayed. In fact, you will distill from the mess of html the data you need, and display or store only that. Multiple times. Faster than you can blink.

The scripting is about to start, in a bit. We need to scope the target first. Rather, we need a target to scope.

- **Step 1: Recon**

Choose what you're going to be scraping. This would be something important to you, which is available on the internet, preferably via the world wide web. This could be across multiple websites, or across many webpages on the same website. Thus you have to separate between scraping a little data over time (say, collecting the top post on the /r/TodayILearned subreddit everyday) or a lot of data at once (say, downloading all the xkcd comics). [Technically reddit has an API for heavier scraping, but one request per day is not going to abuse their system.]

Once identified, the bit of the html element that contains what you need, needs to be identified. In Google Chrome, it's as simple as right-clicking on it and clicking Inspect. Hovering over different parts of the code will also show you the corresponding parts of the page. What you need will be highlighted by default. (Right-click on it and copy its outerHTML or CSS selector, depending on what you're using to do this. You may also need to dig into the nested elements to get to the innermost element that contains the data.) At the end of step one you should know this relevant information of your chosen target. For example, this could be a selector like `"#t3_8zppqc > div._1poyrkZ7g36PawDueRza-J > div > div._1Y6dfr4zLlrygH-FLmr8x- >`